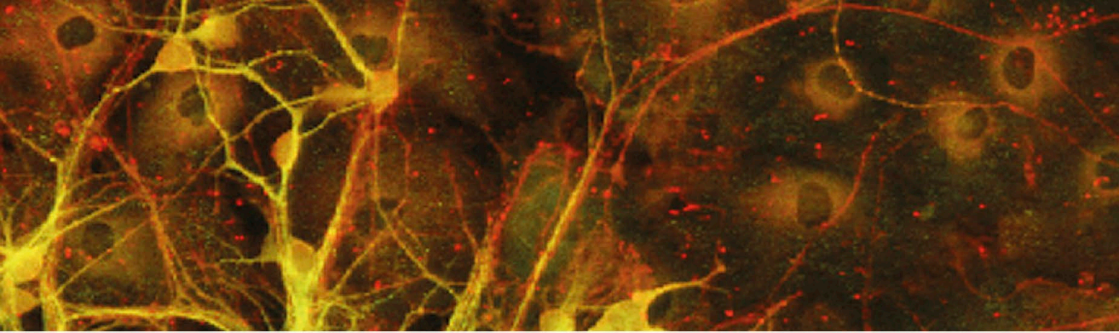




INTRODUCTION

The number of humans in the world has raced past seven billion (7,000,000,000). More than 250 babies are born every minute, while 150,000 people die daily, with the population increasing by almost three humans per second. Each of these people lives and thinks with, and within, that most complex and marvellous of possessions – a human body.

LEVELS OF ORGANIZATION

To understand the inner structure and workings of the human body, this book takes the “living machine” approach, borrowed from sciences such as engineering. This views the body as a series of integrated systems. Each system carries out one major task. In the cardiovascular system, for example, the heart pumps blood through vessels, to supply every body part with essential oxygen and nutrients. The systems are, in turn, composed of main parts known

as organs. For example, the stomach, intestines, and liver are organs of the digestive system. Moving down through the anatomical hierarchy, organs consist of tissues, and tissues are made up of cells.

Cells are often called the building blocks of the body. Active and dynamic, they continually grow and specialize, function, die, and replenish themselves, by the millions every second. The whole body contains about 100 million million cells, of at least 200 different kinds. Science is increasingly able to delve deeper than cells, to the organelles within them, and onwards, to the ultimate components of ordinary matter – molecules and atoms.

ANATOMY

The study of the body's structure, and how its cells, tissues, and organs are assembled, is known as human anatomy. For clarity, its elements are often shown in isolation, because the inside of the body is

